Sun spots — Solution

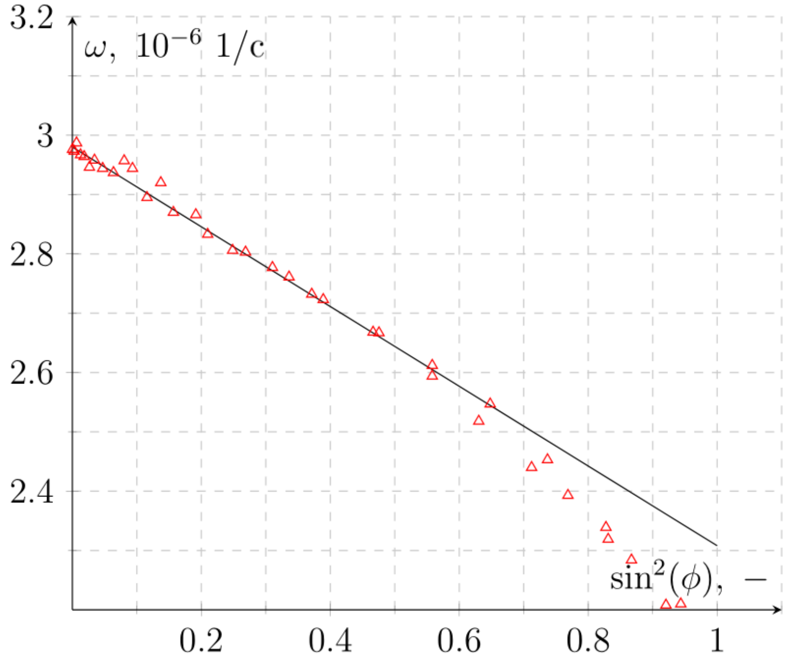
X22 (2022) — English translation

A1 2.00 pts

You are presented with a table of experimental data $\omega(\varphi)$. The data was obtained based on the Doppler effect, which affects the frequency of electromagnetic radiation from the Sun recorded on Earth. Define parameters a and b.

Let's build a graph $\omega(\sin^2\varphi)$. We draw an approximating line along the initial section, from which we find the parameters a and b.

Answer: $a = (2.99 \pm 0.2) \cdot 10^{-6}$ 1/s



A2 0.30 pts

Find the limits of applicability of the formula given above.

From the graph we find the angle $\varphi_{\max}$ after which the points significantly deviate from the straight line.

Answer: $\varphi_{\max} = 50°$

B1 0.30 pts

Obtain all components of the magnetic field on the surface of the Sun $B_{0,r}(\phi)$, $B_{0,\phi}(\phi)$, $B_{0,\lambda}(\phi)$. Express your answers using φ, $\mathfrak{M}$, R_0 and μ_0.

Let us write the field of the magnetic dipole at latitude φ:

$$\vec{B}(\varphi) = \frac{\mu_0}{4\pi}\frac{\mathfrak{M}}{R_0^3}(3\vec{n}\sin\varphi - \vec{z})$$

Projecting this onto the axes corresponding to r, φ and λ we obtain:

Answer:

$$B_{0,r}(\varphi) = \frac{\mu_0}{4\pi}\frac{\mathfrak{M}}{R_0^3}2\sin\varphi$$

B2 0.60 pts

Obtain the magnitude of the magnetic field B_e at the equator inside the Sun near its surface. Express your answer using $\mathfrak{M}$, μ_0, R_0, ξ.

Let's choose a surface stretched over magnetic lines. Then the magnetic field flux through the equatorial section inside the Sun is equal to the flux through one of the hemispheres.

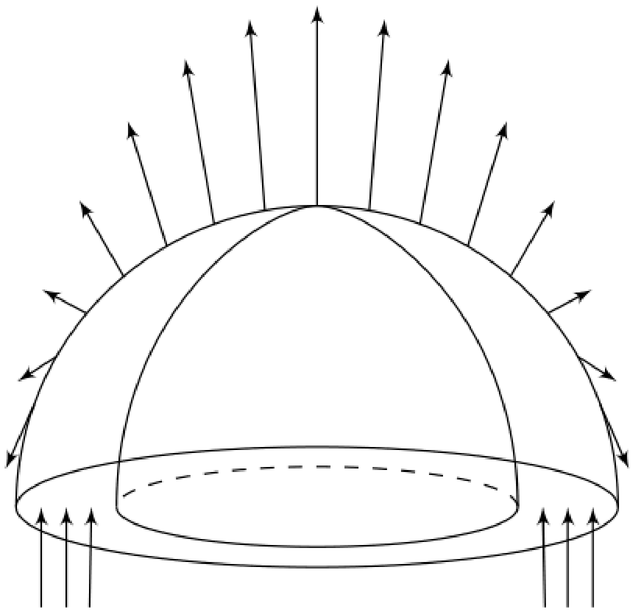
Thus $2\pi R_0 H_0 B_e = \Phi_0$ means $B_e = \frac{\Phi_0}{2\pi\xi R_0^2} = 53$ mT. On the other hand, we can calculate the flux Φ_0 by integrating the dipole field:

$$\Phi_0 = \int\limits_0^{\pi/2} B_{0,r} 2\pi R_0^2 \cos\varphi d\varphi = \frac{\mu_0\mathfrak{M}}{R_0}\int\limits_0^{\pi/2}\sin\varphi\cos\varphi d\varphi = \frac{\mu_0\mathfrak{M}}{2R_0}.$$

Substitute this into the previous formula and get the following:

Answer:

$$B_e = \frac{\mu_0\mathfrak{M}}{4\pi\xi R_0^3}.$$



B3 0.10 pts

Find the numerical value of B_e.

Answer:

$$B_e = \frac{\Phi_0}{2\pi\xi R_0^2} = 53 \text{ mT}$$

B4 — 0.60 pts

Obtain the magnitude of the magnetic field $B_\varphi(\varphi)$ inside the Sun at its surface as a function of latitude φ. Express your answer using φ, B_e.

Now we will also select a surface stretched over magnetic lines. Only now it will extend from the equator to the desired latitude φ.

Then to calculate B_φ you need to find the flux $\Phi(\varphi)$ of the magnetic field of the dipole through the part of latitude from 0 to φ. This is the integral already familiar to us:

$$\Phi(\varphi)(\varphi) = \int\limits_0^\varphi B_{0,r} 2\pi R_0^2 \cos\varphi d\varphi = \frac{\mu_0 \mathfrak{M}}{2R_0}\sin^2\varphi.$$

The flux through the surface is equal to 0:

$$\Phi(\varphi) + 2\pi R_0 \cos\varphi\, H_0 \cos^2\varphi\, B_\varphi(\varphi) = \Phi_0,$$

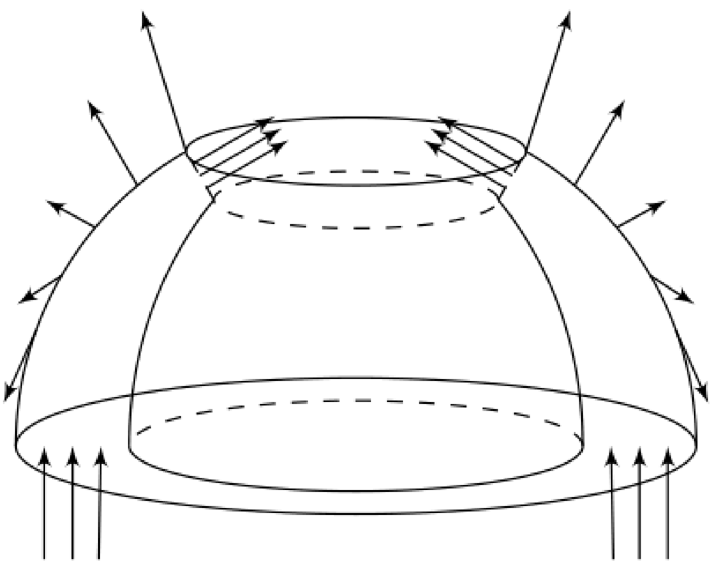
then:

$$2\pi\xi R_0^2 \cos^3\varphi\, B_\varphi(\varphi) = \Phi_0 \cos^2\varphi.$$

So

Answer:

$$B_\varphi = \frac{B_e}{\cos\varphi}.$$



C1 — 1.70 pts

Precisely calculate 30 pairs of points (φ, λ) through which the magnetic line piercing points $\varphi = \pm 50°$, $\lambda = 0$ will pass at time $t = 150$ days.

Let the magnetic line pass through the points $(\pm\varphi_0, \lambda_0)$, where $\varphi_0 = 50°$, $\lambda_0 = 0°$. Then its equation in angular coordinates is given as follows:

$$\lambda = \omega(\varphi)t = (a - b\sin^2\varphi)t$$

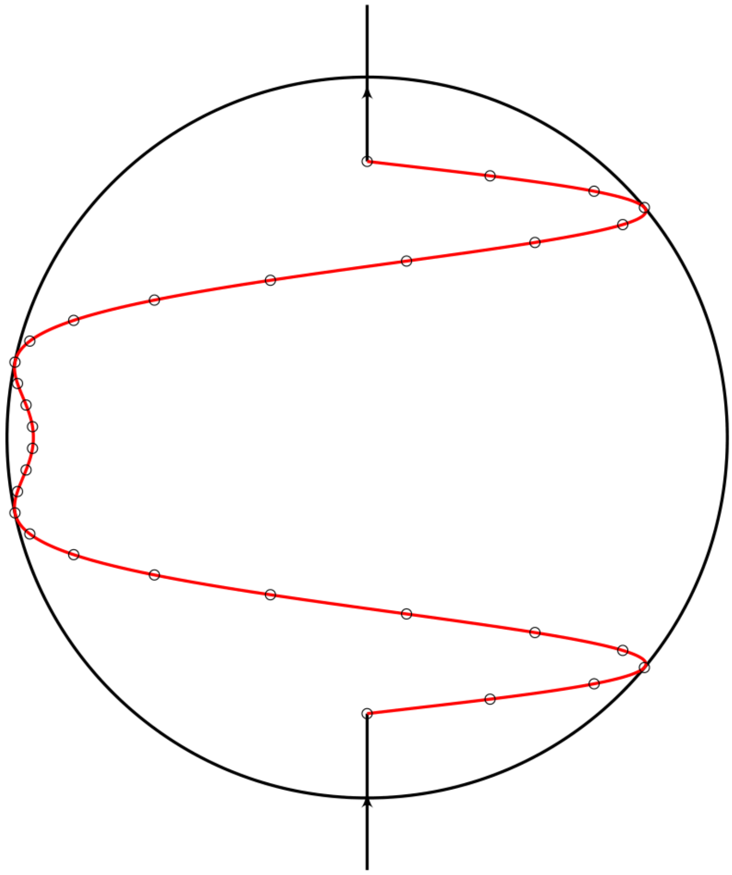
C2 — 1.00 pts

Using this data, plot on your answer sheet what this line will look like for an earthly observer.

Points with angular coordinates in the plane have the following coordinates:

$$\begin{cases} x = R \cdot \sin\lambda \, \cos\varphi \\ y = R \cdot \sin\varphi \end{cases}$$

Let's draw a line through the recalculated points:



C3 — 0.30 pts

Find the angle $\psi(\varphi, t)$ that the magnetic lines make with the meridian. Express your answer using φ, t, a, b.

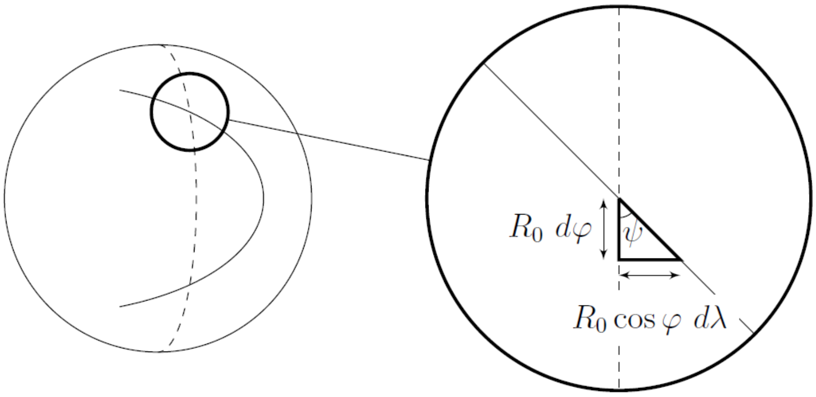
From the picture it is clear that

$$\tan\psi = \frac{d\lambda}{d\varphi}\cos\varphi = 2bt\sin\varphi\cos^2\varphi,$$

is from

Answer:

$$\psi = \arctan\left(2bt\sin\varphi\,\cos^2\varphi\right).$$



C4 — 0.20 pts

Let the magnetic lines inside the Sun near its surface make an angle ψ with the meridian. Find the magnitude of the magnetic field B at this point. Express your answer using ψ, φ, B_e.

The magnetic field flux through the surfaces discussed in Part B should not change, so $B_\varphi = \text{const}$. Then

$$B = \frac{B_\varphi}{\cos\psi}$$

and therefore

Answer:

$$\frac{B_e}{\cos\varphi\ \cos\psi}$$

D1 — 2.70 pts

Using the $\sin\psi \approx 1$ approximation, find B_{crit}.

In the approximation $\sin\psi \approx 1$ it is true that $\tan\psi = 1/\cos\psi$, therefore:

$$B = \frac{B_e \tan\psi}{\cos\varphi} = 2B_e bt \sin\varphi,$$

means the type of curve, $\varphi(t)$, on which spots appear on the Sun is described by the following expression:

$$B_{\text{crit}} = 2B_e bt \sin\varphi,$$

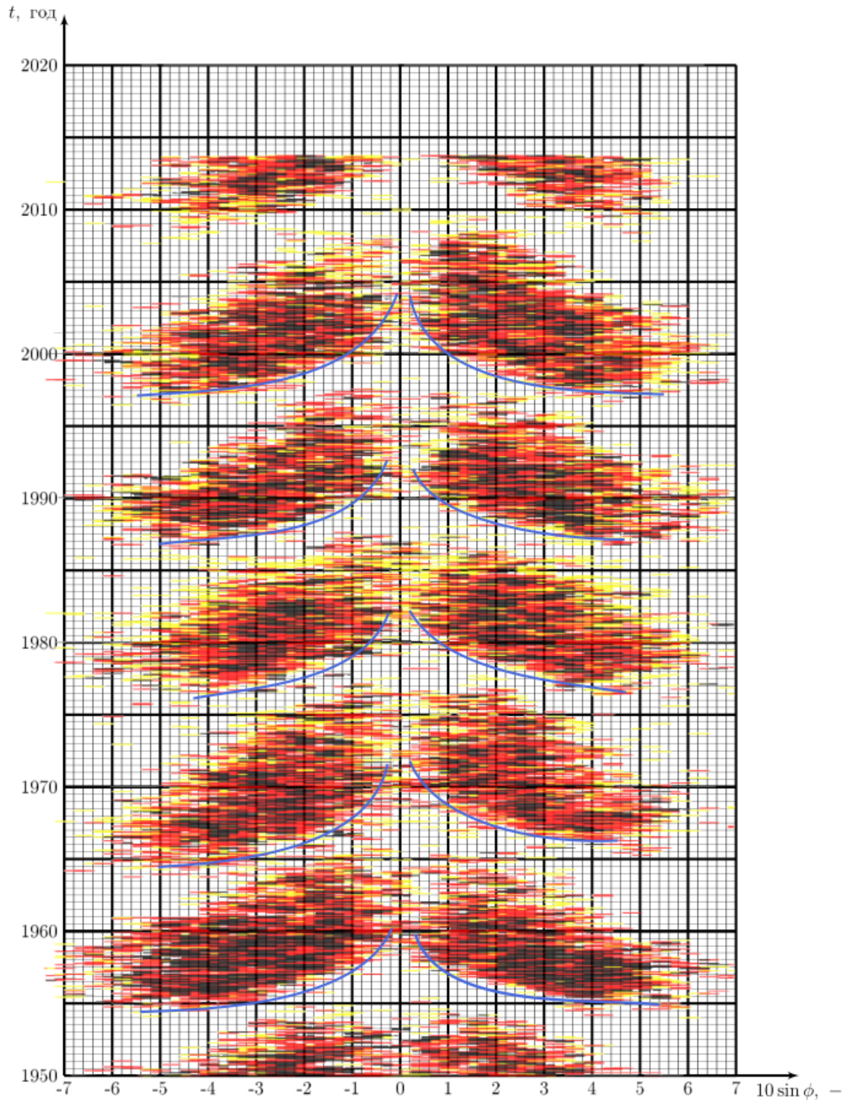
i.e. its linearization is as follows:

$$t = \frac{A}{\sin 2\varphi},$$

where $A = \frac{B_{\text{crit}}}{B_e b}$ and t are counted from the beginning of the 11-year solar cycle (one “butterfly” on the Maunder diagram). By constructing the envelope on the Maunder diagram, we obtain the dependence $t(\sin 2\varphi)$ for critical points (appearance of spots). From it we find A, and then B_{crit}.

Answer:

$$B_{crit} = (15 \pm 5)\,\text{mT}$$



D2 — 0.20 pts

Determine the period of solar activity T_0, i.e. the time after which the Sun's magnetic field returns to its original state.

We find the period as the time between two points with the same "phase" divided by the number of cycles: $T = 11$ year. Next, this number must be increased by 2 times, since after one cycle the field changes direction.

Answer:

$$T_0 = 22\,year$$